# GAGE BALDASSARO


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Junior Developer focused on Gameplay and Graphics Programming

## EDUCATION

**Bachelor of Science in Computer Science | GPA: 3.83/4.0** August 2024 - May 2028
Rice University, Houston, TX
**Courses** Algorithms, Linear Algebra, Multivariable Calculus, Differential Equations, Probability and Statistics, Physics (Mechanics, E&M, Waves, Thermodynamics)

## SKILLS

**Programming Languages** C#, C++, GLSL, Python, Java, MATLAB
**Technical Skills** Unity, Git, OpenGL, ASSIMP, Microsoft Office, JMARS, Cesium, SteamWorks

## PROJECTS

**Code Athens | Lead Programmer** July 2026
*C#, Unity | Top 1% of 10.5k Submissions to GMTK Game Jam 2026 (Narrative, Audio)*

- Collaborated with 5 designers and artists to prototype and ship a polished game within 96 hours.
- Designed an event-driven dialogue engine and a modular, asynchronous loading pipeline for seamless level transitions.
- Rapidly prototyped and developed drag-and-drop minigames and refined based on team feedback and playtesting.

**Mech Combat Demo | Solo Project** May - July 2026
*C#, Unity*

- Prototyped 3rd-person mech combat inspired by Armored Core VI, with a state-driven movement controller for dashing, energy-limited hovering, and boosting.
- Built a custom lock-on camera system with occlusion-aware target searching and velocity-based tilt.
- Designed a ScriptableObject weapon system with quadratic-intercept lead targeting for moving enemies.

**OpenGL Renderer | Solo Project** July - August 2026
*C++, OpenGL, GLSL*

- Engineered a custom multi-pass rendering pipeline from scratch using C++, OpenGL, and GLSL.
- Programmed complex Blinn-Phong lighting and material system using tangent-space matrices for accurate normal mapping.
- Developed configurable post-processing shaders and robust 3D model loading via Assimp to render detailed environments and objects.

## EXPERIENCE

**Unity Developer Intern | *Planetary Parfait*** October 2025 - August 2026
*Experiential Pixels Lab at Rice ECE, Houston, TX*

- Coauthored scientific publication (forthcoming) on immersive visualization in geospatial scientific workflows.
- Engineered procedural terrain integration, collaborating with cross-functional teams to seamlessly render 2D Lunar data from *JMARS* API onto 3D Geospatial Lunar data from *Cesium*.
- Shipped 20+ bug fixes and gameplay features based on community feedback and team playtesting to public Steam build.

**America Reads Mentor** September - December 2025
*Nehemiah Center, Houston, TX*

- Mentored 1st through 4th grade students in literacy and mathematics in afterschool program for students of disadvantaged families in the Houston area.

**Engineering Intern** June - July 2025
*Petrotech Inc., New Orleans, LA*

- Programmed a Human-Machine Interface (HMI) in Excel VBA for monitoring the Programmable Logic Controller of an ALLISON 501-KB Gas Turbine.
- Designed and cleanly documented Modbus TCP/IP communication protocol enabling reliable PLC-to-HMI data exchange via DIGI One IAP.
- Programmed HMI screens based on engineering drawings using FactoryTalk View Machine software.